1 **(a)** $x = 4$ **[B1]**

(b) $y = 3$ **[B1]**

[Total Marks: 2]

2 **(a)** Vertical asymptote: $x = -2$ **[B1]**

Horizontal asymptote: $y = 1$ **[B1]**

(b) Substitute $y = 0$ to find x-intercept: $0 = \dfrac{3}{x+2} + 1 \implies x = -5$ **[M1]**

x-axis at $(-5, 0)$ and y-axis at $(0, 2.5)$ **[A1]**

(c) Two correct branches of the hyperbola sketched **[B1]**

Graph approaches $x = -2$ and $y = 1$ correctly without touching or crossing them **[B1]**

Graph passes through $(-5, 0)$ and $(0, 2.5)$ approximately **[B1]**

[Total Marks: 7]

3 **(a)** $b = 2$ **[B1]**

$c = -1$ **[B1]**

(b) Substitutes $(5, 0)$ into the equation: $\dfrac{a}{5-2} - 1 = 0$ **[M1]**

$a = 3$ **[A1]**

(c) Sets up the equation $\dfrac{3}{x-2} - 1 = -x + 5$ or uses GDC to find intersection of $y = f(x)$ and $y = -x + 5$ **[M1]**

$x = 3$ and $x = 5$ **[A1]**

[Total Marks: 6]